

TECHNICAL AUDIT REPORT

Project: Core Inventory & User Management API

Stack: Python 3.10, FastAPI, PostgreSQL 14

Architecture: Monolithic REST API

Executive Summary

This technical audit identifies raw technical findings only. No prioritization, remediation plan, effort estimation, dependencies, or roadmap are included.

Security

Finding SEC-01

Category	Credentials Management
Location	app/config/settings.py
Evidence	Automated analyzer detected the issue.
Description	Hardcoded DATABASE_PASSWORD and JWT_SECRET detected.

Finding SEC-02

Category	Injection
Location	app/routes/users.py
Evidence	Automated analyzer detected the issue.
Description	Raw SQL built using user input.

Finding SEC-03

Category	CORS
Location	app/main.py
Evidence	Automated analyzer detected the issue.
Description	allow_origins=['*'] configured.

Finding SEC-04

Category	Authentication
Location	app/routes/auth.py
Evidence	Automated analyzer detected the issue.
Description	No rate limiting on login endpoint.

Finding SEC-05

Category	Cryptography
Location	app/utils/crypto.py
Evidence	Automated analyzer detected the issue.
Description	MD5 used for password hashing.

Finding SEC-06

Category	Authorization
----------	---------------

Location	app/routes/admin.py
Evidence	Automated analyzer detected the issue.
Description	Missing role verification on admin endpoint.

Finding SEC-07

Category	HTTP Security
Location	app/main.py
Evidence	Automated analyzer detected the issue.
Description	Security headers not configured.

Database & Performance

Finding DB-01

Category	Indexing
Location	inventory_items
Evidence	Automated analyzer detected the issue.
Description	Missing indexes on frequently filtered columns.

Finding DB-02

Category	ORM
Location	app/routes/inventory.py
Evidence	Automated analyzer detected the issue.
Description	N+1 query pattern detected.

Finding DB-03

Category	Connections
Location	app/database/connection.py
Evidence	Automated analyzer detected the issue.
Description	No connection pooling.

Finding DB-04

Category	Caching
Location	app/routes/analytics.py
Evidence	Automated analyzer detected the issue.
Description	No cache for expensive queries.

Finding DB-05

Category	Pagination
Location	app/routes/orders.py
Evidence	Automated analyzer detected the issue.
Description	Endpoints return unbounded datasets.

Finding DB-06

Category	Transactions
Location	app/services/orders.py
Evidence	Automated analyzer detected the issue.

Description	Missing transaction handling.
-------------	-------------------------------

Code Quality & Architecture

Finding ARCH-01

Category	Architecture
Location	app/routes/monolith_router.py
Evidence	Automated analyzer detected the issue.
Description	Router exceeds 2500 LOC.

Finding CODE-01

Category	Exceptions
Location	app/services/payment.py
Evidence	Automated analyzer detected the issue.
Description	Broad exception handling detected.

Finding CODE-02

Category	Typing
Location	app/routes/orders.py
Evidence	Automated analyzer detected the issue.
Description	Missing type hints.

Finding CODE-03

Category	Validation
Location	Multiple Controllers
Evidence	Automated analyzer detected the issue.
Description	Duplicated validation logic.

Finding CODE-04

Category	Dead Code
Location	app/utils
Evidence	Automated analyzer detected the issue.
Description	Unused modules detected.

Finding CODE-05

Category	Naming
Location	Project Wide
Evidence	Automated analyzer detected the issue.
Description	Inconsistent naming conventions.

Finding CODE-06

Category	Configuration
Location	Project Wide
Evidence	Automated analyzer detected the issue.
Description	Magic numbers and hardcoded constants detected.

Testing & CI/CD

Finding TEST-01

Category	Coverage
Location	tests
Evidence	Automated analyzer detected the issue.
Description	Overall test coverage below 12%.

Finding TEST-02

Category	Isolation
Location	tests/test_shipping.py
Evidence	Automated analyzer detected the issue.
Description	Tests call external services.

Finding TEST-03

Category	Performance Tests
Location	tests
Evidence	Automated analyzer detected the issue.
Description	No load/performance tests.

Finding TEST-04

Category	CI/CD
Location	Project Root
Evidence	Automated analyzer detected the issue.
Description	No CI pipeline configuration detected.

Infrastructure & Deployment

Finding INFRA-01

Category	Dependencies
Location	requirements.txt
Evidence	Automated analyzer detected the issue.
Description	Outdated libraries detected.

Finding INFRA-02

Category	Containerization
Location	Project Root
Evidence	Automated analyzer detected the issue.
Description	Docker configuration missing.

Finding INFRA-03

Category	Filesystem
Location	app/utils/file_storage.py
Evidence	Automated analyzer detected the issue.
Description	Absolute filesystem paths used.

Finding INFRA-04

Category	Environment
----------	-------------

Location	Project Root
Evidence	Automated analyzer detected the issue.
Description	Environment configuration not standardized.

Observability & Documentation

Finding OBS-01

Category	Logging
Location	Project Wide
Evidence	Automated analyzer detected the issue.
Description	Structured logging not implemented.

Finding DOC-01

Category	Documentation
Location	FastAPI Docs
Evidence	Automated analyzer detected the issue.
Description	OpenAPI documentation incomplete.

End of Report

Total Findings: 30

This report intentionally contains only raw findings.